Extended reading — Week 4: TD, SARSA, Q-learning, Expected SARSA

STUDY-RL

May 21, 2026

Reading list

The papers below are the original sources for the algorithms derived in `notes.pdf`. For each: full citation, a one-paragraph summary, one annotated key passage, and the curriculum thread it tightens.

1. Sutton 1988 — "Learning to Predict by the Methods of Temporal Differences"

Citation. Richard S. Sutton, *Machine Learning* 3:9–44, 1988.

Summary. The paper that defined TD methods and proved (informally) their convergence under linear function approximation. Sutton frames TD as the natural compromise between supervised-learning regression on Monte Carlo returns and the dynamic-programming backup. He introduces the λ-return geometric mixture and shows TD(λ) is strictly more sample-efficient than MC on a 5-state random walk.

Why it matters for this week. Every TD algorithm we will see for the next 20 weeks is a descendant of equation (2) of this paper.

> *"Temporal-difference methods are a class of incremental learning procedures specialized for prediction problems. Whereas conventional prediction-learning methods are driven by the error between predicted and actual outcomes, TD methods are similarly driven by the error or difference between temporally successive predictions; with them, learning occurs whenever there is a change in prediction over time."*

Commentary. Note Sutton's emphasis on *change in prediction* as the learning signal. That framing makes TD natural for online deployment: you do not need "ground truth," you only need a stream of predictions you trust to drift toward correctness. Production systems often have exactly this structure — noisy partial outcomes and a stable online estimator.

2. Watkins & Dayan 1992 — "Q-learning"

Citation. Christopher J. C. H. Watkins and Peter Dayan, *Machine Learning* 8:279–292, 1992.

Summary. Defines Q-learning and gives the first convergence proof for off-policy control with a model-free agent. The argument constructs an action-value process driven by a contraction in ℓ_∞ and verifies stochastic-approximation conditions; the behaviour policy can be *anything* that visits each (s, a) infinitely often.

Why it matters for this week. The clearest justification for why we can learn the optimal policy by following a noisy exploratory policy. Foundation for everything in Weeks 5–6 (DQN family).

> *"Q-learning works by directly approximating the action-value function $Q^\star$ — without ever requiring the agent to learn a model of the world or to evaluate alternative policies."*

Commentary. The structural separation between behaviour (what generates data) and target (what we evaluate) is the central abstraction. The deadly triad of Week 5 (off-policy + bootstrapping + function approximation) is born from extending this separation into the function-approximation regime where the convergence proof breaks.

3. van Hasselt 2010 — "Double Q-learning"

Citation. Hado van Hasselt, *NIPS* 2010.

Summary. Identifies and analyses Q-learning's maximisation bias and proposes maintaining two independently updated Q-estimators. Theorem 1 shows that $\mathbb{E}[\max_a \hat{Q}(s,a)] \geq \max_a \mathbb{E}[\hat{Q}(s,a)]$ strictly when there is noise across actions; the gap is upper bounded by the standard deviation of the noise. Empirically demonstrates the fix on a roulette MDP.

Why it matters for this week. Makes the maximisation bias precise. Double-DQN in Week 6 is the function-approximation analogue.

> *"We have shown that the well-known Q-learning algorithm uses the* max *operator to estimate the value of the maximum expected action value, leading to over-estimations of the action values. Double Q-learning, which we introduce in this paper, uses two estimates and decouples the selection from the evaluation, which results in an unbiased value estimator."*

Commentary. The decoupling principle (argmax from one estimator, value from another) reappears in target networks (DQN), twin critics (TD3), and in ensemble disagreement bonuses for exploration. Master it now.

4. van Seijen, van Hasselt, Whiteson, Wiering 2009 — "A Theoretical and Empirical Analysis of Expected Sarsa"

Citation. Harm van Seijen, Hado van Hasselt, Shimon Whiteson, Marco Wiering, *IEEE ADPRL* 2009.

Summary. Defines Expected SARSA, proves it has strictly lower variance than SARSA (same expectation, integrate out next-action noise), and reports faster convergence on cliff-walking-style tasks. Also discusses on- and off-policy variants: when the expectation is taken under the behaviour policy you get on-policy ES; when taken under a target policy you get a different off-policy estimator that subsumes Q-learning (target = greedy) as a special case.

Why it matters for this week. The cleanest example of a free-lunch variance reduction in TD control; the algorithm also unifies SARSA and Q-learning under a single objective.

> *"Expected Sarsa has the same convergence properties as Sarsa, but reduces the variance in the updates and as a result performs better in practice. We argue that this variance reduction is the main reason for the improved performance, and not a difference in bias."*

Commentary. A nice example of Rao–Blackwellisation in RL: replace a sampled estimator by its conditional expectation under a known distribution; never lose bias, always reduce variance. Used implicitly in Expected-SARSA, in soft-Q targets, in GRPO's group-mean baseline, and in many other algorithms we will see later.

5. Sutton & Barto 2018 — *Reinforcement Learning: An Introduction*, Chapter 6

Citation. Richard S. Sutton and Andrew G. Barto, MIT Press, 2nd edition, 2018.

Summary. The textbook chapter that introduces TD prediction and control. The cliff-walking example (Example 6.6) and the maximisation-bias example (Example 6.7) are taken directly from here.

Why it matters for this week. The standard reference; we re-derive its proofs but use its problems verbatim.

> *"TD methods learn a guess from a guess — they bootstrap."*

Commentary. Three words that contain the whole of Weeks 4–10: "learn a guess from a guess" is the entire game. The cost is the deadly triad of Week 5; the benefit is online incremental learning that scales to billions of frames in Week 6.

Sourced from: `research/deeprl_core_sota.md` §1 (DQN family lineage), and the canonical Sutton & Barto textbook for the foundational material.